

Use case name	Generate Shortage Order
Textual Description	The system periodically checks for product shortages. If shortages exist, it queries the Inventory Mock, calculates the best supplier for each missing item, and generates the orders.
List of Actors	Task Scheduler (Primary), Inventory Mock (Secondary)
Pre-conditions	Suppliers and agreements are defined in the system.
Post-conditions	Orders for missing products are saved in the database under the supplier offering the lowest price.
Main success scenario	BEGIN Generate Shortage Order Task Scheduler signals System to check for shortages System requests list of shortages from Inventory Mock Inventory Mock returns list of (productId, missingQuantity) System calculates best supplier for each shortage System generates and saves new orders System logs successful order generation END
Alternatives / Extensions	EXTENSION at step 3: No Shortages IF Inventory Mock returns empty list THEN System logs "No shortages found" EXIT Use Case END IF EXTENSION at step 4: No Supplier Found IF no supplier provides productId THEN System logs error "No supplier found for missing item" System continues to next shortage item END IF